



HEPA Solutions Company

Health Data Analytics | Evidence Generation | Patient Access





Over 40 Projects with our Partners across GCC in 2025!

+25 Centers

Medical Centers, Research Centers, HTA/Economic Departments, and Regulatory Bodies

+50

Companies

Pharm Companies, Local Companies, Medical Devices/Biotech Companies Consulting firms

+40 Projects

Health Economics Model, Budget Impact Report , RWE Generation, Economic Consultation, Class Review Projects, Managed Entry Agreements, Paper and Publication, Education and Advisory Board

+12 Digital Tools

Economic Model, Package Price, Cost Calculator, PROMS, and Digital Market Research



Our Accomplishments and Partnerships across GCC in 2025

+25 Centers

+40 Projects

+50 Companies

+10
Publications

+1500
Experts Database

+12
Digital Calculators



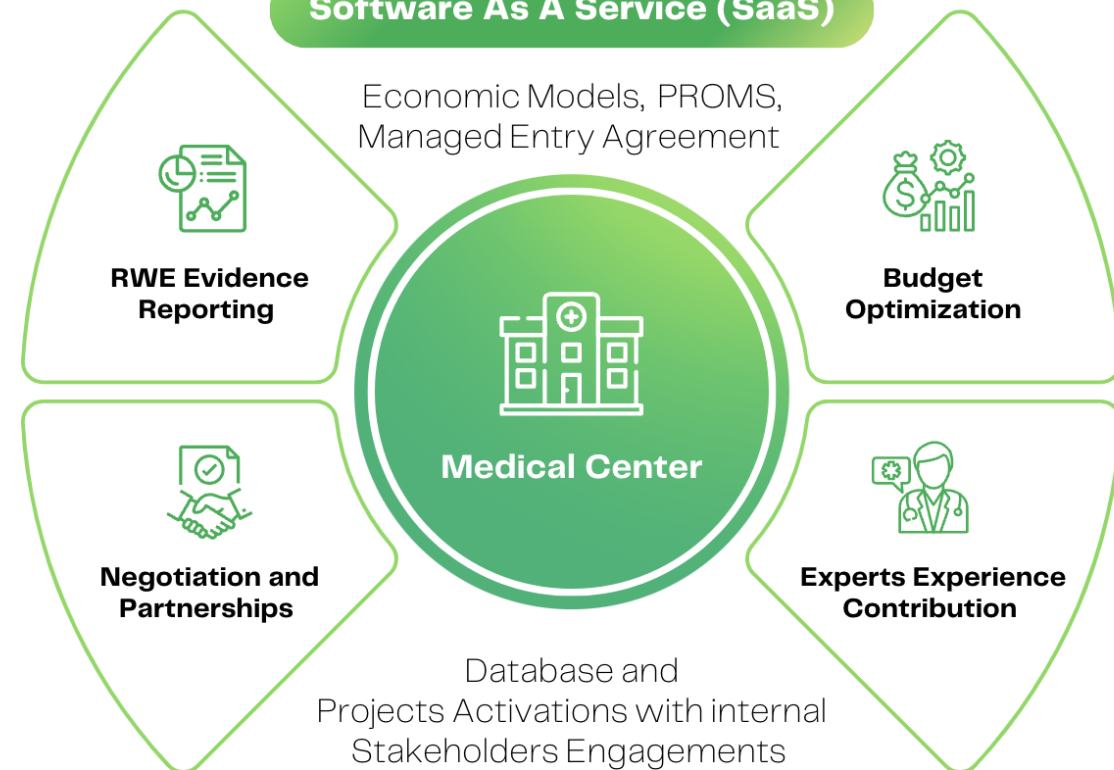
VBHC Platform:

Value Based Analytics and Project Management Platform

Payment Per Service

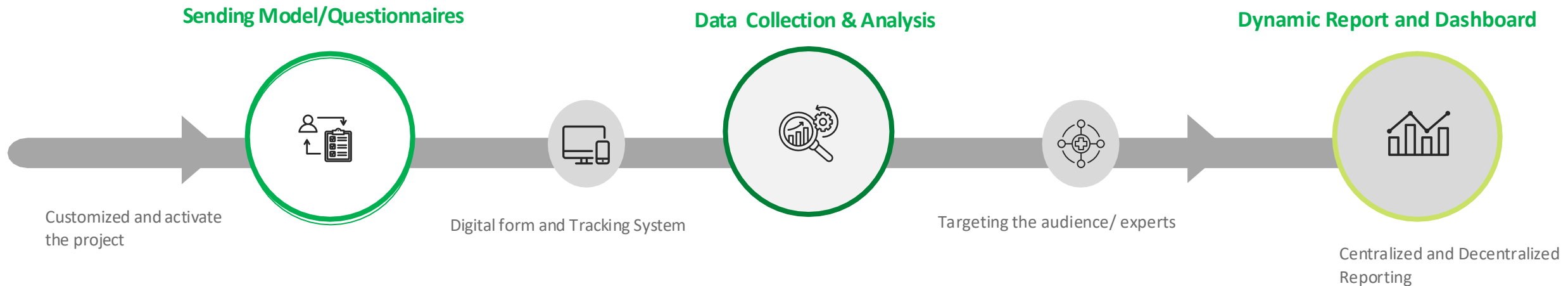
- ▶ Health Economics Model
- ▶ Budget Impact Report
- ▶ RWE Generation
- ▶ Economic Consultation
- ▶ Class Review Projects
- ▶ Managed Entry Agreements
- ▶ Paper and Publication
- ▶ Education and Advisory Board

Software As A Service (SaaS)

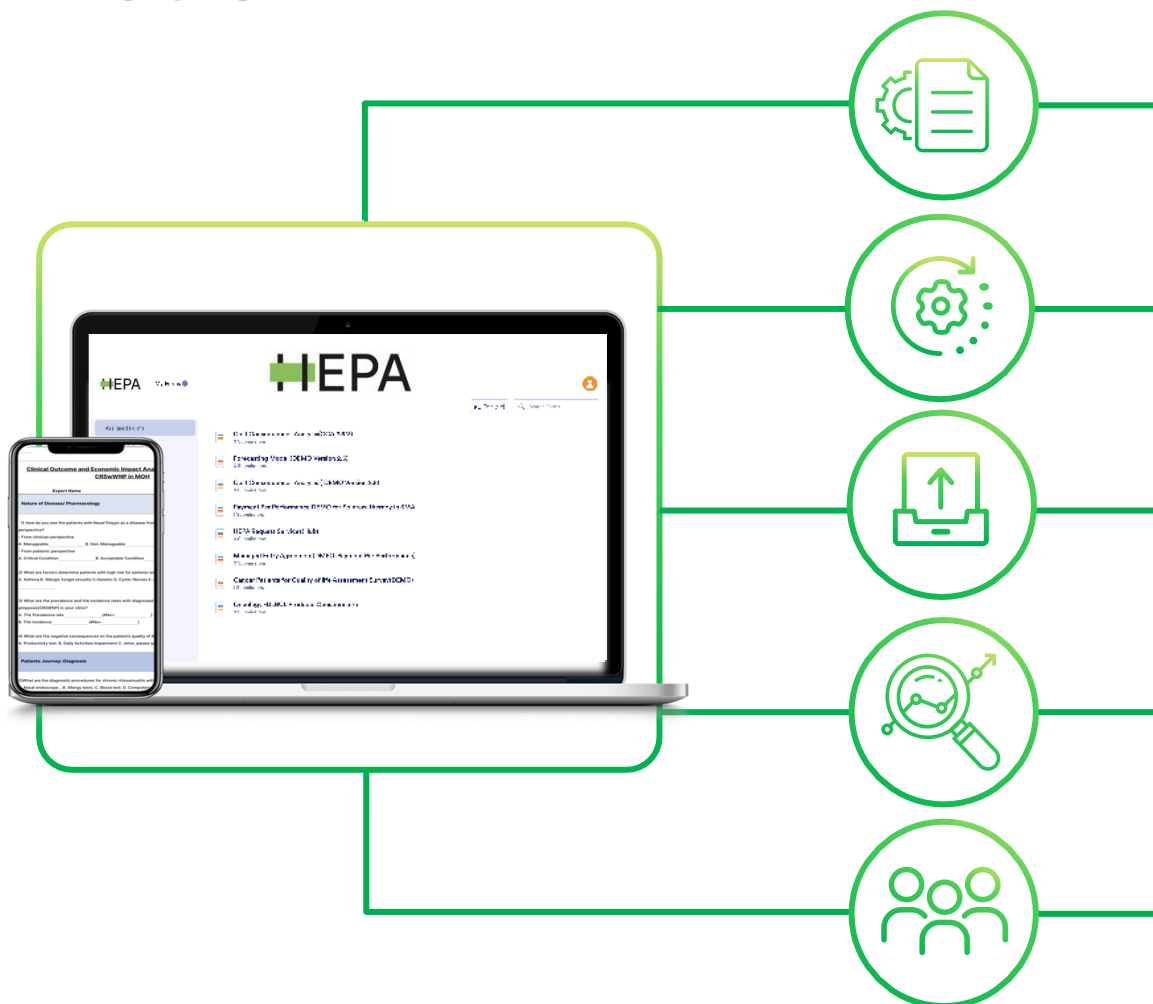


Market Research Platform for Data Collection and Validation

- ▶ Patient Journey Identification
- ▶ Stakeholder Mapping per Therapeutic Area
- ▶ Advisory Board Per Expert
- ▶ Education and Validation Clinical Value
- ▶ Economic Value Perception
- ▶ Payer Insights for Pricing Strategy



Activate Value-Based Healthcare Programs from One Platform



Oncology Budget Optimization Project

Economic models based on built scenario with evidence generation for patient access

Rare and Orphan Drugs Access Project

Identify hidden cost and indirect burden costs for these disease linked to patient assistance program

Value Based Agreement Program

Complex Data Collection and Analysis based on economic model and managed entry agreement

Incentives Patient Program for Obesity and CVD

Medical Package Price Customization based VBA and patient behaviors per country

RWE Generation Based PROMS/PREMS Program

Involve experts and patients with RWE generation linked to VBA



VBAS Assessment with Report Generation

Budget Impact Uncertainty

Low Risk for Reimbursement Impact



Reimbursement Impact On Uncertain Metric

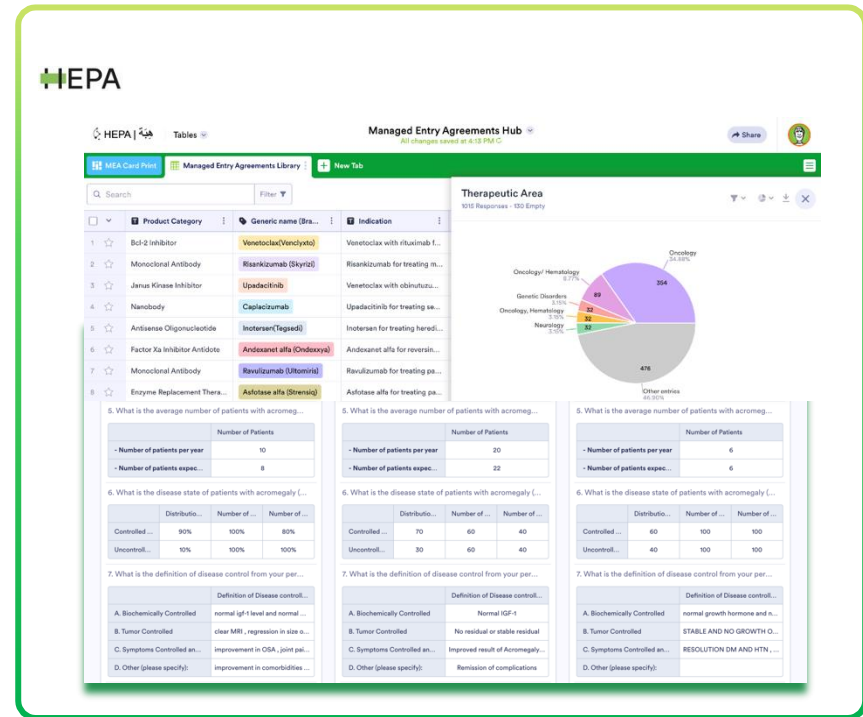
Low Risk with < one High Risk Deal
Medium Risk Equal 2 High Risk Deals
High Risk > 2 High Risk Deals



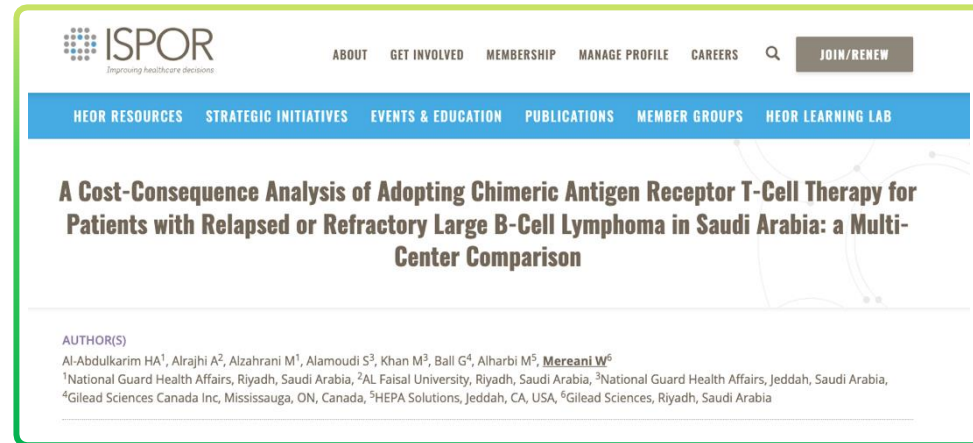
Real World Evidence Generation Customization

Evidence Generation

The image shows a laptop displaying two side-by-side questionnaires from HEPA. The left questionnaire is titled 'Medical Questionnaire' and the right is 'Economic Questionnaire'. Both are for 'Burden of Acromegaly in the Kingdom of Saudi Arabia (KSA)'. The medical questionnaire includes sections for 'Expert Interview-based Structured Questionnaire' and 'Specific clinical categories of interest include the following'. The economic questionnaire includes sections for 'Expert Interview-based Structured Questionnaire' and 'Specific clinical categories of interest include the following'. Both questionnaires have a 'Save' button at the bottom.



Finally, Submit Evidence Generation for Publications!



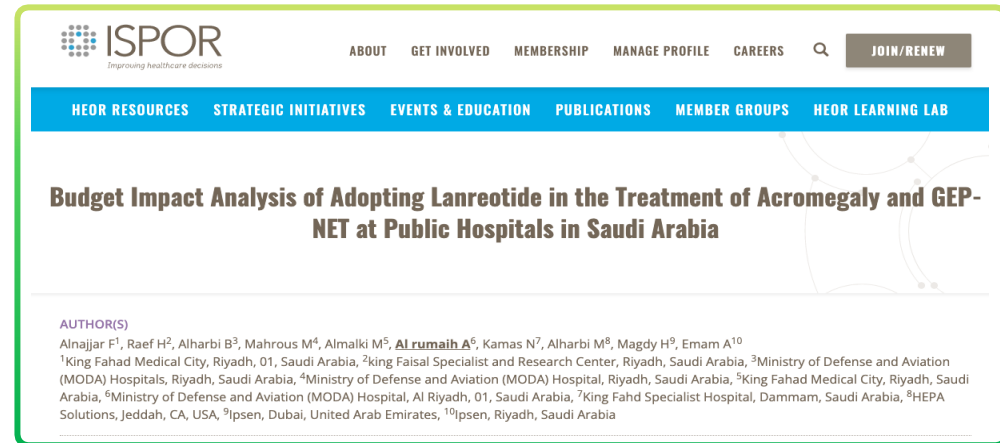
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Improving healthcare decisions

ABOUT GET INVOLVED MEMBERSHIP MANAGE PROFILE CAREERS JOIN/RENEW

HEOR RESOURCES STRATEGIC INITIATIVES EVENTS & EDUCATION PUBLICATIONS MEMBER GROUPS HEOR LEARNING LAB

A Cost-Consequence Analysis of Adopting Chimeric Antigen Receptor T-Cell Therapy for Patients with Relapsed or Refractory Large B-Cell Lymphoma in Saudi Arabia: a Multi-Center Comparison

AUTHOR(S)
Al-Abdulkarim HA¹, Alrajhi A², Alzahrani M¹, Alamoudi S³, Khan M³, Ball G⁴, Alharbi M⁵, **Mereani W⁶**
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Budget Impact Analysis of Adopting Lanreotide in the Treatment of Acromegaly and GEP-NET at Public Hospitals in Saudi Arabia

AUTHOR(S)
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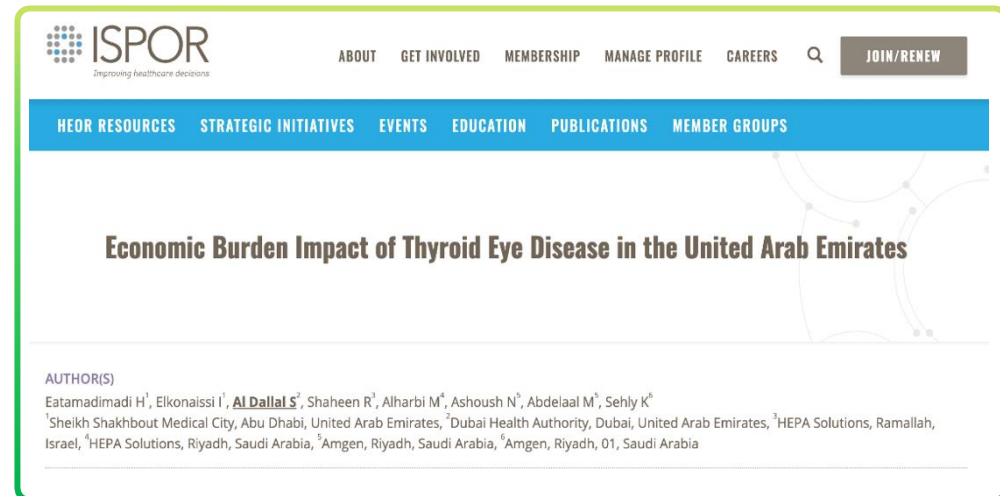
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The Economic Burden of Thyroid Eye Disease in the Kingdom of Saudi Arabia

AUTHOR(S)
Alzahrani A¹, Alabdulkarim H², **Saad Al-Rumaih AM³**, Alayoubi A⁴, Alharbi B⁵, Alhumaidan A⁶, Aljowaisar N⁷, Omaer A⁸, Sumaily A², Alturaiki A⁹, Alharbi I⁹, Alzuman S⁹, Shaheen R¹⁰, Alharbi M¹¹, Ashoush N¹², Abdelaal M¹³, Sehly K¹³
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Health Policy Shaping Case For Economic Impact Study in GCC



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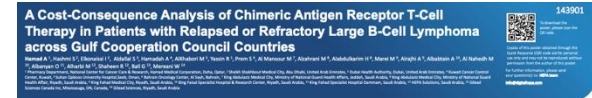
A Cost-Consequence Analysis of Chimeric Antigen Receptor T-Cell Therapy in Patients With Relapsed or Refractory Large B-Cell Lymphoma Across Gulf Cooperation Council Countries

AUTHOR(S)

Hashmi S¹, Elkonaissi I¹, Al dallal S², Hamadah A³, Al-Khabori M⁴, Taha R⁵, **Hamad A**⁶, Prem S⁷, Al-Mansour M⁸, Alzahrani M⁹, Al-Abdulkarim H⁹, Marei M¹⁰, Alrajhi A¹⁰, Albabtain A¹¹, Al Nahedh M¹¹, Albanyan O¹², Alharbi M¹³, Shaheen R¹⁴, Ball G¹⁵, Mereani W¹⁶
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PRESENTATION DOCUMENTS

[CART GCC POSTER for ISPOR 2024143901.pdf](#)



KEY LEARNINGS

Referring CAR T-cell therapy patients to Saudi Arabia (KSA) is a cost-effective alternative for Gulf Cooperation Council (GCC) countries compared to referrals to the United States (US) or Europe.

BACKGROUND

- Relapsed or refractory large B-cell lymphoma (R/R LBCL) is a highly aggressive form of non-Hodgkin lymphoma with limited treatment options, particularly for patients who do not respond to traditional therapies.^{1,2}
- The introduction of CD19 chimeric antigen receptor T-cell (CAR T) therapy offers a promising treatment option with substantial activity against R/R LBCL and a manageable safety profile.^{2,3}
- However, access to CAR T therapy remains a challenge,^{4,5} particularly in regions like the Gulf Cooperation Council (GCC), where there is limited availability of (CAR T) centers. Consequently, CAR T-eligible patients are referred abroad, typically to the United States (US) or Western Europe where more readily accessible treatment centers are available.

OBJECTIVE

- With the availability of CAR T referral centers in the Kingdom of Saudi Arabia (KSA), the aim of this study was to assess the projected economic consequences of adopting (CAR T) referral pathways to the KSA compared to referrals abroad to the US or Western Europe for CAR T-eligible R/R LBCL patients in the GCC.

CONCLUSIONS

- CAR T referral to KSA is a cost-efficient approach for GCC countries compared to referrals to the US or Western Europe.
- Adoption of KSA referrals could potentially enable more patients within the GCC to receive treatment with the same budget spending.

METHODS

- A cost-consequence analysis was conducted to compare the following referral scenarios for CAR T-eligible R/R LBCL patients in the GCC:

Model Scenarios

- Scenario 1: All identified CAR T-eligible patients from GCC countries referred abroad over a 5-year period to the US or Western Europe.
- Scenario 2: Simulation in which all CAR T-eligible patients were referred abroad to KSA instead of to the US or Western Europe. Out of these referrals 50% were to the US and the remaining 50% were to Western Europe.

Model Inputs

- The model incorporated annualized direct medical costs and direct non-medical costs.
- Number of patient referrals, resource utilization and treatment-related costs reported in 2024 USD were sourced from medical bills, public formularies, business centers, and local clinical experts.

Model Outputs

- Annualized per patient and overall costs were calculated to estimate the potential economic impact of the two referral scenarios.
- One-way sensitivity analyses were conducted to account for uncertainty in input parameters and were varied by ± 50%.

RESULTS

- Over a 5-year period, 14 R/R LBCL patients were identified from GCC countries as being referred abroad for CAR T therapy.
- Average per patient costs for referrals abroad to the US or Western Europe were \$1,680,945 including \$516,572 for drug acquisition, \$1,084,375 for direct medical care, and \$52,198 for direct non-medical costs, resulting in total costs of \$2,653,145 for the 14 referred patients.
- Comparable average per patient costs for referral to KSA were estimated at \$636,793 (drug acquisition: \$416,802; direct medical: \$162,642; direct non-medical: \$57,246), which would result in total costs of \$8,914,514 for 14 referred patients.
- With estimated per patient annual cost savings of \$1,043,194, patient referral to the KSA was projected to result in total cost savings of \$14,516,514 representing a 52% reduction in budgetary spend.

Interpretation

- All else considered equal, and despite variations in referral costs across individual GCC countries (Figure 3), utilizing healthcare abroad referrals to KSA could help GCC health authorities realize substantial cost savings for CAR T-eligible patients with R/R LBCL.
- Alternatively, pursuing referrals abroad to KSA instead of the US and Western Europe could possibly lead to a 164% increase in the number of GCC patients receiving CAR T therapy for the same budget spending.

Figure 1. Model Structure

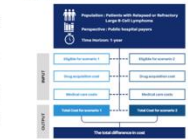


Table 1. Assumptions per patient about referrals

Cost Component	CAR T Referral to US/Europe	CAR T Referral to KSA	Referral to US/Europe
Drug Acquisition	\$516,572	\$416,802	\$516,572
Medical Care	\$1,084,375	\$162,642	\$1,084,375
Non-medical Care	\$52,198	\$57,246	\$52,198
Total Cost	\$1,680,945	\$636,793	\$2,653,145

Figure 2. Annualized total costs per patient for referral abroad to the US or Western Europe

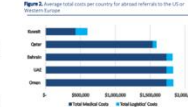


Figure 3. The patient and overall cost comparison of CAR T therapy between referral to the US or Western Europe



Figure 4. Annual cost breakdown per patient



Figure 5. Annual cost breakdown for 14 patients



Economic Burden and Impact of Disease for KSA and UAE



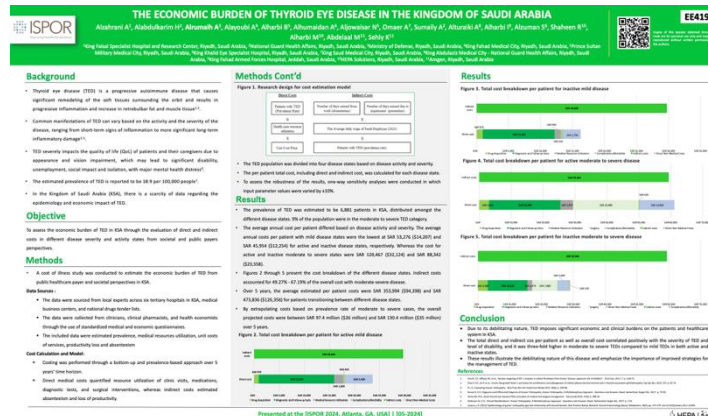
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EE457 ECONOMIC BURDEN IMPACT OF THYROID EYE DISEASE IN THE UNITED ARAB EMIRATES

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¹Sheikh Shakhboub Medical City, Abu Dhabi, United Arab Emirates, ²Dubai Health Authority, Dubai, United Arab Emirates, ³HEPA Solutions, Riyadh, Saudi Arabia, ⁴Amgen, Riyadh, Saudi Arabia, ⁵Amgen, Riyadh, 01, Saudi Arabia

Objectives: Thyroid eye disease (TED) is a rare debilitating disease associated with significant morbidity. The aim of this study was to assess the cost of illness of TED in the United Arab Emirates (UAE). **Methods:** A cost of illness model was developed to evaluate the economic burden of TED over a 1-5 years' time horizon. The model employed a bottom-up, prevalence-based approach. Clinical and economic data were collected from local experts through targeted structured questionnaires included direct and indirect costs from public payers and societal perspectives. Direct medical costs were retrieved from national insurance for outpatient visits, medications, laboratory tests, imaging, and surgery costs. Indirect costs included absenteeism and productivity loss based on the experts' inputs. Through extrapolation of cost and prevalence data, the economic burden was calculated. **Results:** With an estimated prevalence of 1500 patients in the UAE distributed among active mild to moderate disease, moderate to severe disease, and very severe cases (Sight threatening). Disease severity had varying levels of impact on total cost directly and indirectly. The annual costs per patient was AED 102,156, AED 244,188, and AED 260,587 (active mild to moderate, moderate to severe disease, and very severe cases, respectively). Over 5 years with diseases transition, average costs per patient was AED 959,145 with moderate to severe disease. Indirect costs accounted over 56.3-59.5% of the overall costs from the total patients with active moderate to severe and severe to very severe cases. By extrapolating costs based on prevalence rate of moderate to severe cases, the overall projected costs were between AED 428.9 million and AED 564.1 million over 5 years. **Conclusions:** This is the first study assessing the economic burden of TED in the UAE. The findings postulate the considerable disease burden and provide valuable insights into key cost drivers.



Reference Published Economic Projects

 The Economic Burden of Thyroid Eye Disease **in the Kingdom of Saudi Arabia**

Link: <https://lnkd.in/dYQy3Gcv>

 Economic Burden Impact of Thyroid Eye Disease **in the United Arab Emirates**

Link: <https://lnkd.in/dr6BR6dC>

 A Cost-Consequence Analysis of Adopting Chimeric Antigen Receptor T-Cell Therapy for Patients with Relapsed or Refractory Large B-Cell Lymphoma **in Saudi Arabia**

Link: [https://www.valueinhealthjournal.com/article/S1098-3015\(23\)00456-4/fulltext](https://www.valueinhealthjournal.com/article/S1098-3015(23)00456-4/fulltext)

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Link: https://www.ispor.org/docs/default-source/euro2024/cart-gcc-poster-for-ispor-2024143901-pdf.pdf?sfvrsn=5c241772_0

 Economic Impact of Adopting Lanreotide for Patients with Acromegaly and Gep-Net **in Qatar**

Link: <https://www.ispor.org/heor-resources/presentations-database/presentation/euro2024-4015/143928>

 The Economic Burden of Acromegaly Disease in the Kingdom of Saudi Arabia

Link: <https://www.ispor.org/heor-resources/presentations-database/presentation-cti/ispor-europe-2025/poster-session-5-2/the-economic-burden-of-acromegaly-disease-in-the-kingdom-of-saudi-arabia>

 Cost of Illness Study: Cushing's Syndrome in the Kingdom of Saudi Arabia

Link : <https://www.ispor.org/heor-resources/presentations-database/presentation-cti/ispor-europe-2025/poster-session-2-2/cost-of-illness-study-cushing-s-syndrome-in-the-kingdom-of-saudi-arabia>

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Health Data Analytics | Evidence Generation | Patient Access

